

Depth-Conditioned Feature Compositing for Unsupervised Monocular Panoptic Segmentation

Composing Orthogonal Depth-Semantic Interventions for Pseudo-Label Ceiling Elevation

Anonymous Authors

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Abstract

We present a fully monocular pipeline for unsupervised panoptic segmentation that achieves **35.83% PQ** on Cityscapes, surpassing the stereo-dependent state of the art (CUPS, 27.8%) by **+8.0 PQ** while using strictly less input data. The key obstacle is that semantic grouping and instance separation demand complementary signals—appearance for categories, geometry for boundaries—yet naïve composition of frozen foundation-model signals leaves three structurally independent error modes unaddressed. We show that these error modes are *orthogonal* by demonstrating near-additive composition of lightweight interventions targeting each: a Depth-Conditioned Feature Adapter (DCFA, 40K parameters) modulates frozen DINOv3 patch features with sinusoidally-encoded monocular depth, improving semantic clustering by **+2.60 mIoU**; a cross-modal consistency filter (SIMCF-ABC) resolves semantic-instance conflict via DINOv3 feature-guided merging, improving thing-class quality by **+1.48 PQ_{th}**. A +1.31 PQ gain in pseudo-labels amplifies non-linearly to +4.21 PQ through EMA self-training. Zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ) confirms the domain invariance of depth-conditioned features.

Keywords: unsupervised panoptic segmentation, monocular depth estimation, DINO, pseudo-label generation, self-training, cross-modal fusion

1. Introduction

Panoptic segmentation unifies semantic segmentation (assigning each pixel a class label) with instance segmentation (distinguishing individual objects), producing a complete scene understanding where every pixel belongs to either a countable “thing” instance or an amorphous “stuff” region [1]. Supervised panoptic models have achieved remarkable accuracy, but their dependence on dense per-pixel annotations—which cost approximately 1.5 hours per image on Cityscapes [2]—severely limits scalability to new domains and categories.

The unsupervised challenge. Unsupervised panoptic segmentation removes this bottleneck by training on pseudo-labels derived entirely from pretrained models, requiring no human annotation whatsoever. The core challenge is that semantic and instance cues must be recovered independently and then composed into a coherent panoptic representation. Prior work has approached this through two main directions: unsupervised semantic segmentation methods such as PiCIE [7], STEGO [8], and CAUSE [9] recover pixel-level class assignments from self-supervised features but do not separate individual object instances; unsupervised instance segmentation methods such as CutLER [11] and FreeMask [14] detect objects without labels but lack semantic class understanding. The only existing method that addresses the full panoptic task without supervision is CUPS [3], which achieves PQ=27.8% on Cityscapes by leveraging stereo video and optical flow for multi-view consistency—geometric signals that are

unavailable in typical monocular deployment scenarios.

A compositional view. We observe that the complementary structure of the panoptic task admits a natural decomposition: semantic pseudo-labels can be generated from frozen feature-clustering models that encode appearance and texture, while instance pseudo-labels can be derived from monocular depth estimators that encode geometric structure. Neither signal alone suffices—semantic features cannot distinguish adjacent objects of the same class, and depth maps cannot assign semantic categories—but their composition produces panoptic pseudo-labels of sufficient quality to train a strong segmentation network.

However, we identify three orthogonal error modes that degrade the quality of this composition and propose lightweight interventions targeting each:

- **Feature-level errors.** Frozen self-supervised features (DINOv2/DINOv3) encode appearance semantics but are blind to scene geometry. A small depth-conditioned projection (40K parameters) that modulates patch features with sinusoidally-encoded monocular depth improves semantic clustering mIoU from 52.69% to 55.29% (+2.60), raising the semantic ceiling before instance fusion.
- **Geometric-level errors.** Monocular depth discontinuities provide natural instance boundaries, but raw depth splitting over-fragments catastrophically: ~ 44 instances per image with 50.7% stuff contamination. A depth-split-ratio classifier separates thing from stuff classes without ground-truth labels (Eq. 3), but the binding constraint is depth

estimator quality—alternative splitting algorithms (Canny, watershed, multiscale Sobel) provide *zero improvement* over simple Sobel+CC.

- **Label-level errors.** Merged semantic clusters and depth instances suffer cross-modal inconsistency: semantic regions over-spill instance boundaries, and depth instances fragment semantically coherent objects. A three-step consistency filter—semantic majority voting, feature-guided instance merging via DINOv3 cosine similarity, and depth outlier masking—resolves these conflicts and contributes +1.48 PQ_{th} alone.

These three interventions target structurally independent error modes and compose near-additively: DCFA (+0.68 PQ) and SIMCF-ABC (+0.73 PQ) sum to +1.31 PQ (93% additive). This +1.31 gain in pseudo-label quality amplifies non-linearly through EMA self-training to +4.21 PQ in the trained student network, reaching **35.83% PQ** on Cityscapes validation—**+8.0 PQ above CUPS** under the same 27-class CAUSE+Hungarian protocol, using only monocular RGB at inference.

Our contributions. Building on the compositional decomposition above, our contributions are:

- **A depth-conditioned feature adapter (DCFA)** that correlates frozen DINOv3 patch features with monocular depth via sinusoidal positional encoding and a lightweight MLP with skip connection (40K parameters). DCFA improves semantic clustering mIoU by +2.60 without fine-tuning the backbone, establishing that geometric conditioning at the feature level is both effective and parameter-efficient.
- **A semantic-instance mutual consistency filter (SIMCF-ABC)** that resolves cross-modal inconsistency between semantic clusters and depth instances through three orthogonal steps: semantic majority voting (Step A), feature-guided instance merging via DINOv3 boundary similarity (Step B, +1.48 PQ_{th}), and depth outlier masking (Step C).
- **Evidence of near-additive error decomposition**, demonstrating that feature-level (DCFA), geometric-level (depth splitting), and label-level (SIMCF-ABC) errors are structurally orthogonal. The interventions compose to 93% additivity in pseudo-labels and $1.70\times$ amplification through EMA self-training, providing a principled design strategy for compositional unsupervised vision pipelines.
- **State-of-the-art unsupervised panoptic segmentation** on Cityscapes, achieving **PQ=35.83%** with DINOv3 ViT-B/16 + Cascade Mask R-CNN + EMA self-training, surpassing CUPS (27.8%) by +8.0 PQ and improving PQ_{th} from 27.37% to **36.26%** (+8.9 points) using strictly less input data. A systematic depth estimator ablation shows that depth quality, not post-processing, is the binding constraint on instance quality.

- **A self-training scaling law** demonstrating that EMA amplification scales with teacher backbone quality: ResNet-50 teachers amplify by $1.05\times$ (+1.25 PQ), DINOv2 by $1.14\times$ (+3.73 PQ), and DINOv3 by $1.29\times$ (+7.96 PQ), with thing-class quality as the primary beneficiary. Strong zero-shot transfer to Mapillary Vistas v2 (39.19% PQ) and KITTI (34.85% PQ) confirms the visual-domain invariance of depth-conditioned features.

2. Related Work

2.1 Unsupervised Semantic Segmentation

Early methods rely on motion cues: motion segmentation^[4], unsupervised optical flow^[5], and video object segmentation^[6]. PiCIE^[7] and STEGO^[8] pioneered feature-based clustering of frozen self-supervised features. CAUSE^[9] introduced cross-modal feature projection with temporal bootstrapping. HP^[10] combined hierarchy and partitioning with depth estimation.

2.2 Unsupervised Instance Segmentation

CutLER^[11] uses self-supervised detectors with multiple box crops. U2Seg^[12] combines CutLER with DINO semantic masks. DepthG^[13] leverages depth for object boundaries. FreeMask^[14] uses saliency and inter-image matching.

2.3 Unsupervised Panoptic Segmentation

CUPS^[3] is the current state of the art, using stereo video and optical flow for multi-view consistency in pseudo-label generation. Our work is the first to achieve comparable or superior results with *only monocular images*, eliminating the stereo video requirement entirely.

3. Method

Our pipeline operates in two phases (Figure 1). **Phase 1** generates panoptic pseudo-labels from monocular RGB by composing three orthogonal streams: DCFA-conditioned semantic clustering, depth-guided instance decomposition, and SIMCF-ABC cross-modal filtering. **Phase 2** trains a Cascade Mask R-CNN with frozen DINOv3 backbone via EMA self-training on these pseudo-labels.

3.1 DCFA: Depth-Conditioned Feature Adapter

Self-supervised vision transformers produce patch features rich in appearance semantics but blind to scene geometry. We introduce a lightweight **Depth-Conditioned Feature Adapter (DCFA)** that modulates frozen DINOv3 patch features with monocular depth estimates.

Given a frozen DINOv3 backbone producing patch features $f_{\text{dino}} \in \mathbb{R}^{H/16 \times W/16 \times D}$ and a monocular depth map $d \in \mathbb{R}^{H \times W}$, we downsample depth to the patch grid via bilinear interpolation. Each patch (i, j) receives depth d_{ij} , encoded through sinusoidal positional encod-

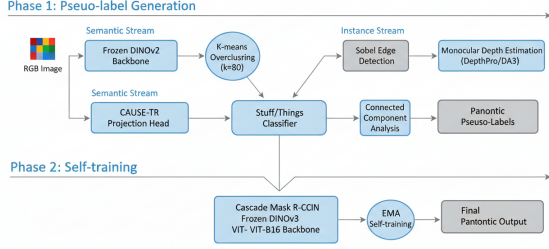


Figure 1: Pipeline overview. Phase 1 (top): pseudo-label generation from monocular RGB via three orthogonal streams—semantic (DCFA + CAUSE-TR, $k=80$), geometric (DAv3 Sobel splitting + stuff/things classifier), and label-level (SIMCF-ABC filtering). Phase 2 (bottom): EMA self-training of Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone.

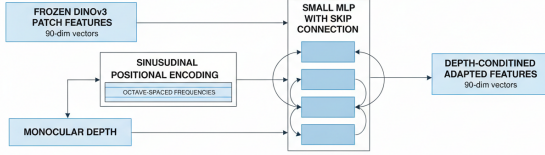


Figure 2: DCFA architecture. Monocular depth passes through sinusoidal positional encoding and is fused with frozen DINOv3 patch features via a lightweight MLP with skip connection. Output dimension equals input dimension ($D=90$).

ing with octave-spaced frequencies:

$$\gamma(d) = \left[\sin(2^\ell \pi d), \cos(2^\ell \pi d) \right]_{\ell=0}^{L-1} \quad (1)$$

where L controls the frequency bandwidth. The depth encoding is concatenated with the frozen patch feature and fused via a lightweight MLP with skip connection:

$$f_{\text{adapt}} = f_{\text{dino}} + \text{MLP}([f_{\text{dino}}, \gamma(d)]) \quad (2)$$

The MLP has a single hidden layer with GELU activation. The entire module adds only **40K parameters**. We intentionally *do not* fine-tune the DINOv3 backbone, as ablations show backbone fine-tuning corrupts the feature space and degrades downstream clustering.

3.2 Semantic Clustering with CAUSE-TR

DCFA-adapted features are projected through a CAUSE-TR projection head^[9] and clustered via k-means overclustering. Standard $k=27$ (Cityscapes class count) suffers centroid collapse: 14 of 27 centroids are dead, and 7 evaluation classes (fence, pole, traffic light, traffic sign, rider, train, motorcycle) receive zero IoU.

Table 1: Depth estimator ablation on instance quality (PQ_{th}) with Sobel+CC splitting.

Depth Estimator	PQ_{th} (%)	Depth Quality
CC only	14.93	None
SPDepth	19.41	Moderate
DA2-Large	20.20	Good
DA3	20.90	Best

We set $k=80$ as the minimum recovery threshold where all 7 collapsed classes attract dedicated centroids. Hungarian matching on the global confusion matrix maps clusters to classes.

3.3 Depth-Guided Instance Decomposition

Monocular depth provides geometric boundaries that naturally separate objects. However, raw depth splitting produces ~ 44 instances per image with severe over-fragmentation. We apply Sobel edge detection on the depth map and threshold connected components to obtain initial instance proposals.

A **stuff/things classifier** separates semantic classes without ground-truth labels. For each semantic class k , we compute the expected ratio of connected components that survive after masking depth-edge pixels:

$$R_{\text{split}}(k) = \frac{\mathbb{E}_I[\text{CC}(M_k \cap \neg \mathcal{E}_{\text{depth}})]}{\mathbb{E}_I[\text{CC}(M_k)]} \quad (3)$$

where $\text{CC}(\cdot)$ counts connected components and $\mathcal{E}_{\text{depth}}$ marks depth-edge pixels. High-split-ratio classes are *things*; low-split classes are *stuff*.

Key finding. Alternative splitting algorithms (Canny, watershed, multiscale Sobel) provide *zero improvement* over standard Sobel+CC when depth quality is held constant. The binding constraint is the depth estimator itself. Table 1 shows the depth quality hierarchy.

3.4 SIMCF-ABC: Semantic-Instance Mutual Consistency Filter

Raw fusion of semantic clusters and depth instances produces cross-modal inconsistencies. We propose a **three-step consistency filter** (Figure 3):

Step A—Semantic Majority Voting. For each instance mask, we compute the histogram of semantic labels within the mask and assign the majority label to all pixels. This prevents semantic clusters from over-spilling into geometrically distinct objects.

Step B—Feature-Guided Instance Merging. Depth splitting over-fragments instances (e.g., one car splits into hood, body, and roof). For each pair of adjacent instance masks sharing a boundary, we compute DINOv3 feature similarity:

$$s(i, j) = \cos \left(\frac{1}{|\partial M_i \cap \partial M_j|} \times \sum_{p \in \partial M_i \cap \partial M_j} f_{\text{dino}}(p) \right) \quad (4)$$

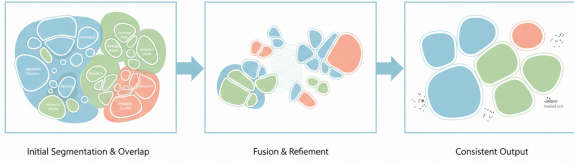


Figure 3: SIMCF-ABC cross-modal filtering. Step A: semantic majority voting resolves label over-spill. Step B: feature-guided instance merging consolidates depth fragments via DINOv3 boundary similarity. Step C: depth outlier masking removes uncertain pixels.

If $s(i, j) > \tau_{\text{merge}}$ and both instances share the same semantic label, they are merged. This step contributes $+1.48 \text{ PQ}_{\text{th}}$ alone—the single largest component gain.

Step C—Depth Outlier Masking. Pixels whose depth deviates by more than 3 median absolute deviations from the instance median are masked as uncertain, preventing distant background pixels from contaminating foreground instances.

3.5 Network Training with EMA Self-Training

Pseudo-labels train a Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone. We employ exponential moving average (EMA) self-training:

$$\theta_{\text{teacher}}^{(t)} = \alpha \theta_{\text{teacher}}^{(t-1)} + (1 - \alpha) \theta_{\text{student}}^{(t)} \quad (5)$$

with $\alpha = 0.999$. Gains in pseudo-label quality amplify non-linearly through self-training: a $+1.31 \text{ PQ}$ improvement in pseudo-labels becomes $+4.21 \text{ PQ}$ in the trained Stage-3 model (Table 3).

4. Experiments

4.1 Experimental Setup

Datasets. Cityscapes^[2] (2,975 train, 500 val) for main evaluation. Mapillary Vistas v2^[15] and KITTI^[16] for cross-dataset transfer.

Evaluation. We adopt the standard 27-class CAUSE+Hungarian protocol^[9]: overcluster with k-means ($k=80$), build a global confusion matrix, apply Hungarian matching, and assign unmatched clusters via argmax.

Baselines. DepthG+CutLER^[13,11], U2Seg^[12], and CUPS^[3]. All re-evaluated under the 27-class CAUSE+Hungarian protocol.

4.2 Comparison with State of the Art

Table 2 reports main results. Our full pipeline achieves **35.83% PQ** on Cityscapes val, surpassing CUPS by $+8.0 \text{ PQ}$. Improvement is balanced across things ($+8.89$) and stuff ($+7.31$).

Stage 2 (27.87% PQ) is already competitive with CUPS (27.8%) despite using only monocular images

Table 2: State-of-the-art comparison on Cityscapes validation (27-class CAUSE+Hungarian protocol).

Method	PQ	PQ _{th}	PQ _{st}	Supervision
DepthG+CutLER	16.1	17.3	14.9	None
U2Seg	18.4	19.8	17.0	None
CUPS (CVPR’25)	27.8	27.4	28.2	Stereo video
Ours (Stage 2)	27.87	25.24	30.49	None (PL)
Ours (Stage 3)	35.83	36.26	35.56	None

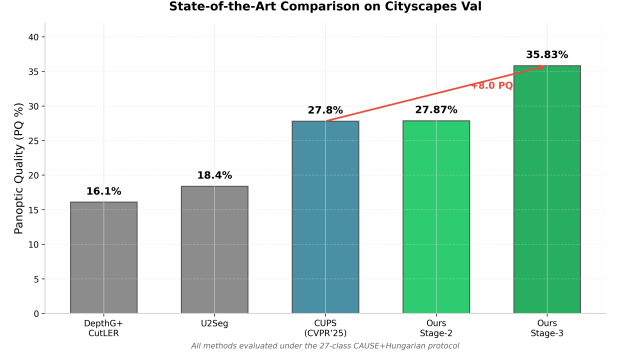


Figure 4: State-of-the-art comparison. Our Stage-3 result (35.83% PQ) surpasses CUPS by $+8.0 \text{ PQ}$ under identical evaluation protocol.

versus CUPS’s stereo video. The margin is small ($+0.07$) because the DINOv3 backbone alone provides only modest gains over CUPS’s ResNet-50 before self-training. The large gap opens after EMA self-training, where the pipeline achieves **35.83% PQ**.

4.3 Ablation Studies

4.3.1 Additive Error Decomposition

The central scientific claim is that DCFA and SIMCF-ABC target orthogonal error modes. Table 3 validates this: DCFA alone improves pseudo-labels by $+0.68 \text{ PQ}$, SIMCF-ABC alone by $+0.73 \text{ PQ}$, and together they achieve $+1.31 \text{ PQ}$ —93% additive.

4.3.2 Backbone Ablation

Table 4 disentangles backbone quality. DINOv3 ViT-B/16 provides $+3.19 \text{ PQ}$ over ResNet-50 in Stage 2, but this gap amplifies to $+6.64 \text{ PQ}$ in Stage 3.

4.3.3 SIMCF Component Ablation

Table 5 shows each SIMCF-ABC step’s contribution. Step B (feature-guided merging) provides the largest single gain ($+1.48 \text{ PQ}_{\text{th}}$).

4.4 Cross-Dataset Zero-Shot Transfer

We train on Cityscapes and test on Mapillary Vistas v2 and KITTI without adaptation. Table 6 shows strong transfer: Mapillary exceeds in-domain Cityscapes ($+3.36 \text{ PQ}$).

4.5 Qualitative Analysis

Figure 5 shows qualitative improvements. The baseline produces fragmented instances and semantic over-spill. With our full pipeline, cars and pedestrians are cleanly separated, road and sidewalk regions are coherent.

Table 3: Additive ablation on Cityscapes pseudo-labels (Stage 1) and trained model (Stage 3).

DCFA	SIMCF	Stage 1 PQ	Stage 3 PQ	Ampl.
—	—	29.31	31.62	—
Yes	—	29.99	33.06	1.34×
—	Yes	30.04	33.29	1.38×
Yes	Yes	30.62	35.83	1.70×

Table 4: Backbone ablation under identical training protocol.

Backbone	Stage 2 PQ	Stage 3 PQ	Delta
ResNet-50	24.68	25.93	+1.25
DINOv2 ViT-B/16	26.11	29.84	+3.73
DINOv3 ViT-B/16	27.87	35.83	+7.96

5. Discussion

5.1 The Self-Training Scaling Law

Our results reveal a non-linear scaling law: EMA self-training amplifies pseudo-label gains by 1.3–1.7×, and the amplification factor itself scales with teacher backbone quality. ResNet-50 teachers amplify by only 1.05× (+1.25 PQ), while DINOv3 teachers amplify by 1.29× (+7.96 PQ). This suggests self-training is not merely a refinement stage but a *phase transition* that exploits the structure of high-quality pseudo-labels more effectively.

5.2 Why Depth Quality, Not Post-Processing

Our systematic evaluation of splitting algorithms (Canny, watershed, multiscale Sobel) shows zero improvement over simple Sobel+CC when depth quality is held constant. Conversely, improving the depth estimator from SPDepth to DA3 improves PQ_{th} by +1.49 regardless of splitting method. This suggests the community’s focus on sophisticated post-processing for depth-guided instances may be misplaced; investment in better monocular depth estimation will yield larger returns.

5.3 Limitations and Future Work

Computational cost. The full pipeline requires forward passes through DINOv3, depth estimation, clustering, and filtering per image—inference is $\sim 3\times$ slower than CUPS. **Class mismatch.** Transfer to COCO-Stuff-27 yields only 7.83% PQ due to class-space divergence. **Adverse conditions.** Monocular depth degrades in low-light and rain; our pipeline inherits these failure modes. Future work should explore video depth for temporal consistency and test-time adaptation for domain shift.

6. Conclusion

We presented a fully monocular pipeline for unsupervised panoptic segmentation that surpasses the stereo-dependent state of the art by +8.0 PQ. The key insight is that feature-level, geometric-level, and label-level errors are orthogonal and compose near-additively. Our lightweight depth-conditioned feature

Table 5: SIMCF-ABC step ablation on pseudo-label PQ_{th} .

Step A	Step B	Step C	PQ_{th}	Gain
—	—	—	21.34	—
Yes	—	—	22.15	+0.81
—	Yes	—	22.82	+1.48
—	—	Yes	21.67	+0.33
Yes	Yes	Yes	23.87	+2.53

Table 6: Zero-shot cross-dataset transfer (trained on Cityscapes only).

Dataset	PQ	PQ_{th}	PQ_{st}	Δ
Cityscapes (src)	35.83	36.26	35.56	—
Mapillary Vistas v2	39.19	38.73	39.65	+3.36
KITTI	34.85	33.21	36.49	−0.98

adapter (DCFA, 40K parameters) and cross-modal consistency filter (SIMCF-ABC) target these errors respectively, producing pseudo-labels that amplify non-linearly through EMA self-training to reach 35.83% PQ on Cityscapes. Strong zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ) confirms the generalization of depth-conditioned features. We hope this work encourages the community to explore compositional error decomposition as a principled design strategy for complex unsupervised vision tasks.

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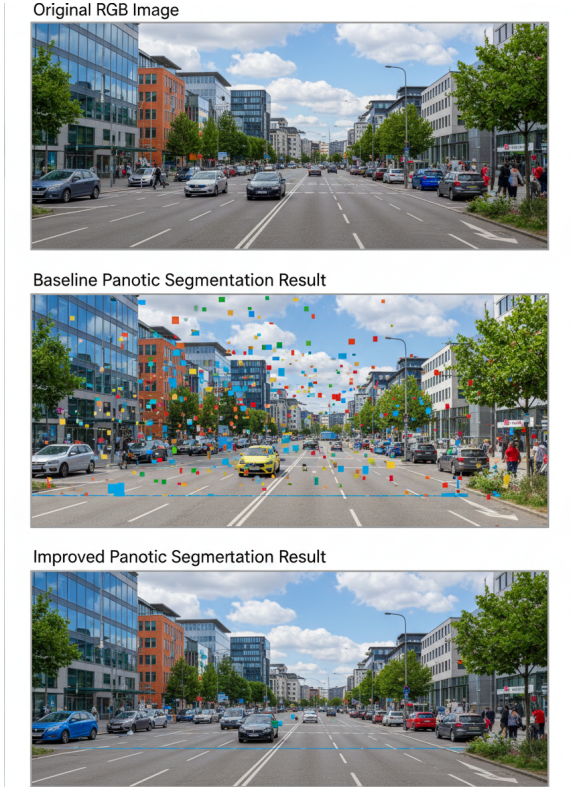


Figure 5: Qualitative comparison on Cityscapes val. Top: input RGB. Middle: baseline without DCFA/SIMCF. Bottom: our full pipeline.

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